

Population Dynamics in Asian Countries: Trends in Fertility, Mortality, and Life Expectancy

Bonnie Bao (2609757) James Teng (2609563)
Yunshi Zhou (2645618) Stan Xu (2644192)
Lyuluo Xie (2609629)

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1 Introduction

Population dynamics are closely connected to economic development, healthcare systems, and long-term social planning. In recent decades, many Asian countries have experienced declining fertility rates, increasing life expectancy, and lower child mortality. These demographic changes affect labor markets, education systems, and public policy.

This project examines three major indicators from the World Bank World Development Indicators database: life expectancy at birth, under-5 mortality rate, and adolescent fertility rate. Our research question is: **How have major population health indicators changed across selected Asian countries over time, and what regional patterns can be observed?**

Overall, we find that most countries experienced substantial improvements in life expectancy and child survival, while fertility rates declined significantly. However, progress was uneven across countries.

2 Data Description

2.1 Data Source

The dataset was obtained from the World Bank World Development Indicators (WDI) database. We selected countries located in Asia and focused on yearly observations from 2000 to 2023.

2.2 Indicators Used

- **Life expectancy at birth (years)** — SP.DYN.LE00.IN
- **Mortality rate, under-5 (per 1,000 live births)** — SH.DYN.MORT
- **Adolescent fertility rate (births per 1,000 women ages 15–19)** — SP.ADO.TFRT

2.3 Data Cleaning and Pre-processing

Several steps were performed before analysis:

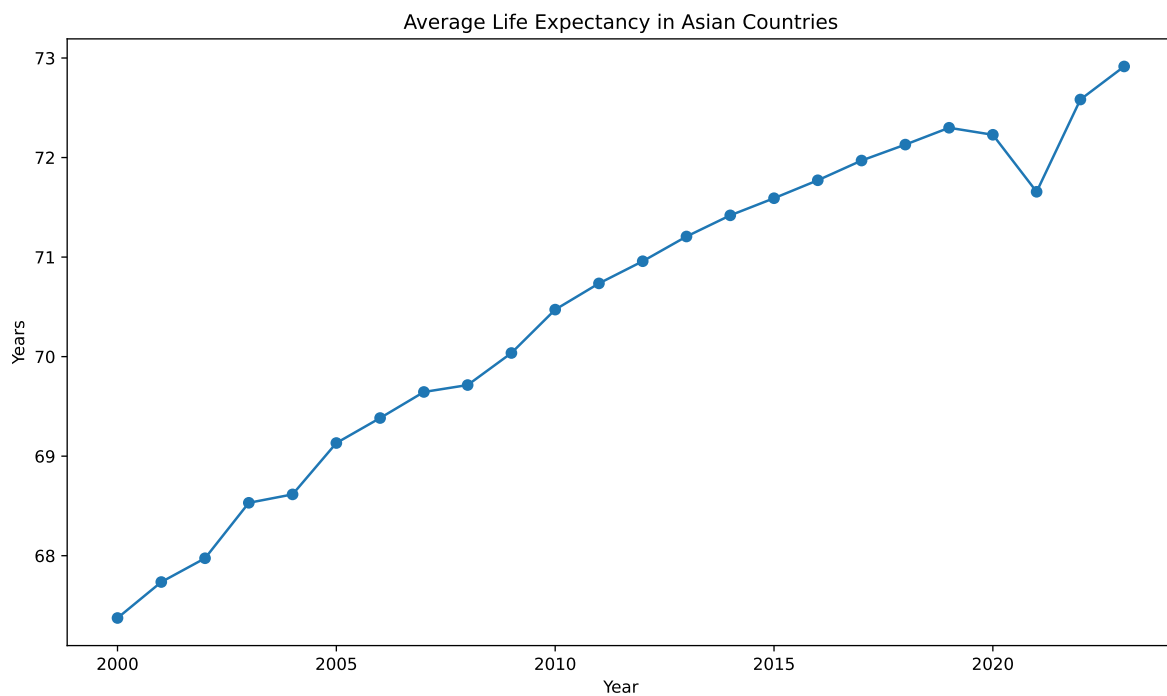
1. Filtered observations to Asian countries only.
2. Selected years from 2000 to 2023.
3. Removed duplicate rows.
4. Converted year and indicator values into numeric format.
5. Addressed missing values by excluding incomplete observations in visual analyses.
6. Merged indicators by country and year.

2.4 Key Variables

- `country` : country name
- `year` : observation year
- `life_expectancy` : years at birth
- `mortality_u5` : deaths per 1,000 live births
- `fertility_adolescent` : births per 1,000 women ages 15–19

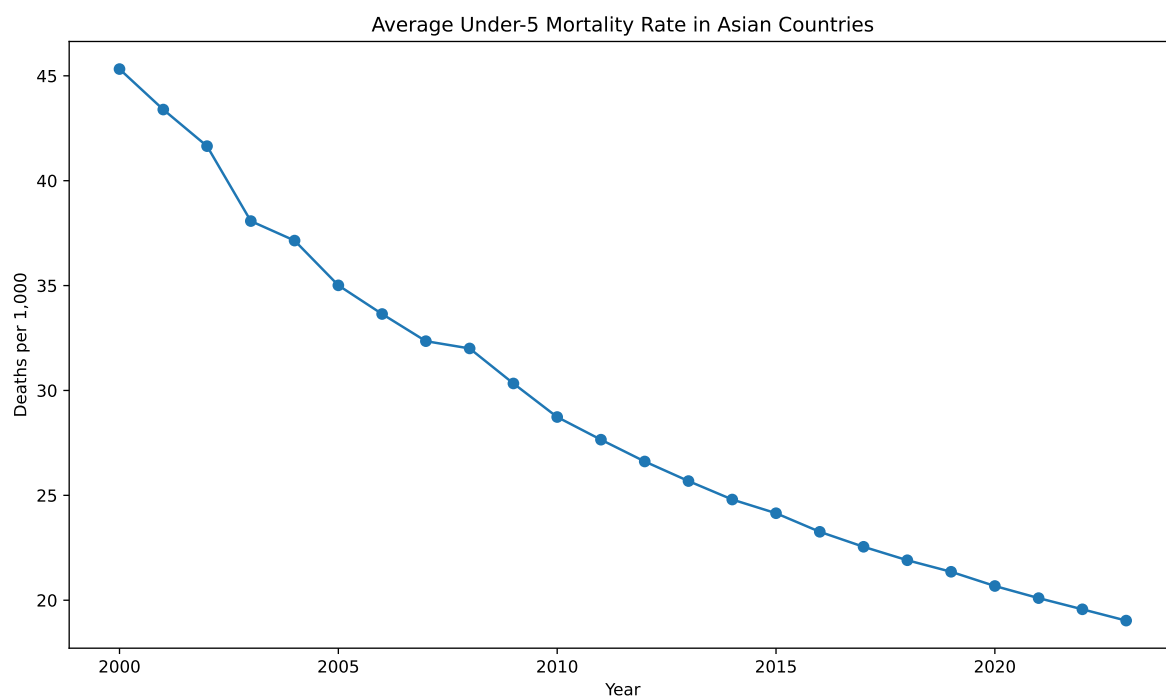
3 Data Analysis

3.1 Trend in Life Expectancy



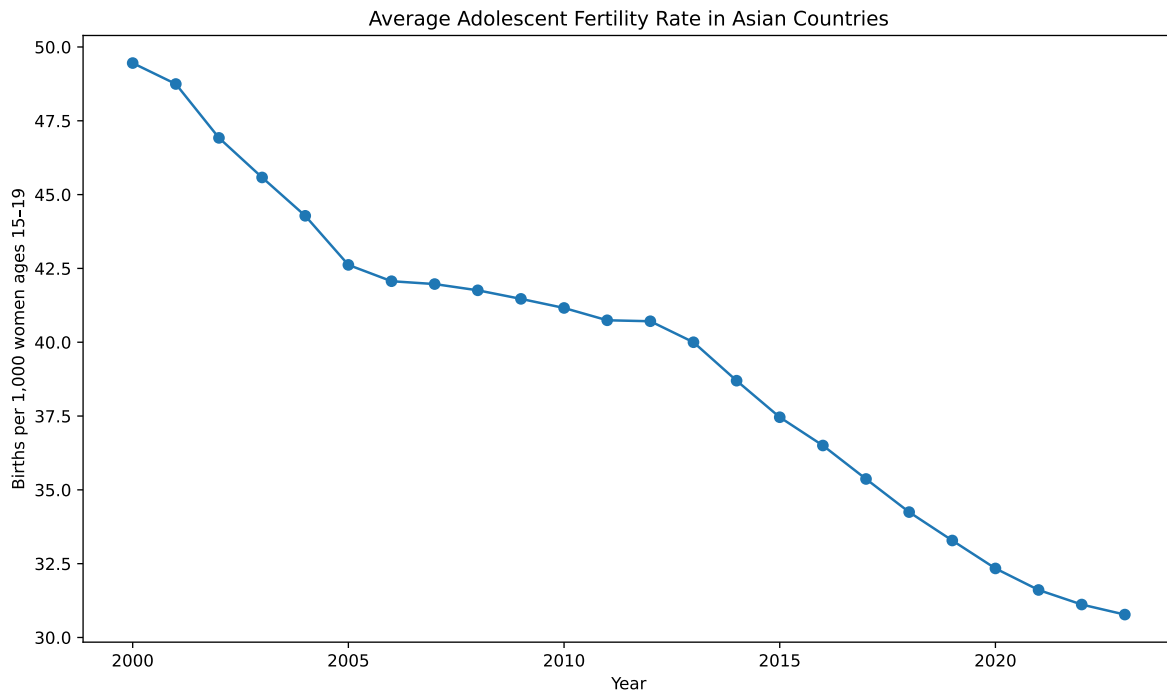
Average life expectancy steadily increased over the study period, reflecting medical progress, improved nutrition, and broader healthcare access.

3.2 Trend in Under-5 Mortality



Child mortality declined sharply across the region, suggesting strong improvements in maternal care, vaccination coverage, and sanitation.

3.3 Trend in Adolescent Fertility



Most countries showed declining adolescent fertility, which may reflect rising female education levels and delayed marriage.

4 Results and Discussion

4.1 Life Expectancy

The results reveal a broad and consistent upward trend in life expectancy across Asia between 2000 and 2023. The regional average rose by several years over this period, reflecting widespread improvements in healthcare infrastructure, disease control, and living standards. Countries with stronger economic growth and more developed healthcare systems — particularly high-income East Asian economies such as Japan, South Korea, and Singapore — maintained the highest life expectancy throughout the study period, consistently above 80 years by 2023.

Among lower-income countries, the gains were even more notable in relative terms. Bangladesh, Bhutan, and Nepal all recorded substantial increases from comparatively low baselines, suggesting that targeted public health investments can produce meaningful demographic progress even without high per capita income. A notable feature of the data

is a small dip in life expectancy around 2020–2021 visible in several countries, most likely attributable to excess mortality during the COVID-19 pandemic. Recovery was observed in most countries by 2022–2023, indicating the resilience of health systems in the region.

4.2 Under-5 Mortality

Child mortality declined sharply and consistently across all countries in the dataset. This decline is one of the most striking demographic achievements of the past two decades and reflects improvements in a range of interconnected factors: expanded vaccination programs, better access to clean water and sanitation, improved maternal nutrition, and reductions in communicable disease burden.

Countries that entered the study period with the highest child mortality rates — including Cambodia, Myanmar, and several South Asian nations — recorded the steepest absolute declines, narrowing the gap with more developed economies. This convergence pattern suggests that international health initiatives and domestic policy reforms have been effective in targeting the most vulnerable populations. Despite this progress, inequality persists: as of 2023, child mortality in some lower-income countries remained several times higher than in high-income East Asian economies, pointing to ongoing challenges in healthcare access and infrastructure.

4.3 Adolescent Fertility

Adolescent fertility declined across most countries over the study period, though the pace and magnitude of change varied considerably. The sharpest reductions were observed in countries that simultaneously expanded girls’ secondary education and strengthened family planning services. Rising female educational attainment is widely regarded as one of the strongest predictors of declining adolescent fertility, as education delays marriage, raises awareness of reproductive health, and expands economic opportunities for young women.

In contrast, some countries with limited educational infrastructure or persistent cultural barriers to girls’ schooling showed slower progress. This divergence reinforces the view that adolescent fertility reduction is not automatic with economic growth alone — it requires deliberate social and policy investment. The overall downward trend nonetheless suggests that meaningful change is achievable across diverse country contexts.

4.4 Regional Patterns and Demographic Transition

Taken together, the three indicators paint a picture of an Asia-Pacific region in the midst of a broad demographic transition. According to demographic transition theory, societies progress from a state of high birth and death rates to one characterised by low fertility and low mortality as development proceeds. The data presented here are broadly consistent with this framework:

countries at higher levels of economic development display lower fertility and mortality and higher life expectancy, while less developed countries are progressing along the same trajectory at varying speeds.

The most developed East Asian economies appear to be in a late stage of this transition, with life expectancy plateauing at very high levels and adolescent fertility approaching zero. Several South and Southeast Asian nations are at earlier stages, still recording meaningful improvements across all three indicators. This heterogeneity within the region underscores the importance of context-specific policies rather than one-size-fits-all approaches to population health.

4.5 Limitations

This analysis has several limitations worth acknowledging. First, some countries had missing observations for certain years, which may introduce gaps in trend analysis and limit the comparability of findings across countries. Second, national averages conceal substantial internal regional inequality — urban and rural populations within the same country may experience very different demographic realities. Third, the analysis is primarily descriptive and does not establish causal relationships between policy interventions and demographic outcomes. Fourth, the three indicators selected, while informative, represent only a subset of the factors that shape population dynamics; incorporating additional variables such as maternal mortality, healthcare expenditure, or educational attainment would provide a more complete picture.

5 Conclusion

This project examined three major population health indicators — life expectancy at birth, under-5 mortality rate, and adolescent fertility rate — across Asian countries from 2000 to 2023 using data from the World Bank’s World Development Indicators database. The analysis drew on SQL-based descriptive statistics and Python visualisations to identify trends, cross-country comparisons, and regional patterns.

The findings are unambiguous in direction: life expectancy increased substantially across all countries in the dataset, while under-5 mortality and adolescent fertility declined. These shifts are not minor statistical fluctuations but represent profound improvements in human welfare. Millions of children who would previously have died before their fifth birthday now survive; millions of adolescent girls are delaying childbearing and staying in education longer. These are among the most consequential demographic changes a society can undergo.

The pace and magnitude of progress, however, varied considerably across countries. High-income East Asian economies such as Japan and South Korea achieved very high levels of population health early in the period and maintained them, while lower-income countries in South and Southeast Asia recorded faster rates of improvement from lower starting points.

This convergence dynamic is encouraging and suggests that the gap between the region's most and least developed nations is narrowing, though it has not closed. Countries such as Bangladesh and Bhutan stand out as success stories of rapid demographic improvement driven by sustained investment in healthcare and female education.

The strong co-movement of all three indicators — confirmed by their high pairwise correlations — highlights the interconnected nature of population health. Improvements in child survival tend to coincide with increases in life expectancy and declines in adolescent fertility because they share common underlying drivers: access to healthcare, sanitation, education, and economic opportunity. This suggests that policies targeting any one of these dimensions are likely to generate positive spillovers across the others, making comprehensive approaches to development particularly effective.

Future research could extend this analysis in several directions. Comparing sub-regions within Asia — for example, East Asia versus South Asia, or coastal versus landlocked countries — would reveal whether geographic and structural factors explain the divergence in outcomes. Incorporating panel regression models would allow researchers to quantify the separate contributions of income growth, healthcare spending, and educational attainment to demographic change. Longer time horizons extending back to 1960 would also enable a fuller assessment of the demographic transition in countries that began the process earlier. Finally, linking these population indicators to labour market and economic growth data would shed light on the long-run consequences of demographic change for development in the region.